

Predictive Interaction Field Theory: Deriving the Interaction Source Tensor from Expected Free Energy and Information-Theoretic Action

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Abstract

This paper establishes a definitive link between Bayesian mechanics and Riemannian geometry by formalizing Interaction Field Theory (IFT) through the lens of Expected Free Energy (EFE). We move beyond reactive models of behavior by deriving the Interaction Source Tensor ($T_{\mu\nu}$) from a variational principle applied to an action functional where the Lagrangian is identified with the expectation of future surprisal. We prove that $T_{\mu\nu}$ naturally decomposes into gradients of Ambiguity and Relative Risk, providing a rigorous physical basis for goal-directed behavior. The resulting non-linear field equations, $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, describe a self-organizing manifold where behavioral geodesics emerge from the minimization of predictive uncertainty. This work provides a unified generative model for complex adaptive systems, offering a structural solution to the exploration-exploitation trade-off in behavioral science.

1 Introduction

The formal description of how sentient agents interact with their environment has traditionally been bifurcated into two distinct domains: the probabilistic inference models of cognitive science and the dynamical systems of classical physics. However, recent advances in Information Geometry and the Free Energy Principle (FEP) suggest a deeper, underlying symmetry between the way systems represent information and the way they move through state-space. This paper aims to close this gap by proposing a Predictive Interaction Field Theory (PIFT), where the dynamics of interaction are governed by the same geometric principles that describe gravitation in general relativity.

At the core of our proposal is the recognition that every autonomous system must minimize its Expected Free Energy (EFE) to maintain structural integrity and achieve goal-directed behavior (Friston et al., 2023). While previous works have treated this minimization as a gradient flow on a statistical manifold, we argue that this process is fundamentally a geometric phenomenon. By transitioning from a purely time-dependent interaction rule to

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a state-dependent one, $\gamma = \gamma(t, I_S)$, we induce a Riemannian manifold where the "distance" between interaction states is defined by the agent's predictive uncertainty.

The primary contribution of this work is the rigorous derivation of the Interaction Source Tensor ($T_{\mu\nu}$). We move beyond the descriptive catalogs of heuristics found in behavioral economics by demonstrating that $T_{\mu\nu}$ emerges naturally from an action functional where the "matter" of the system is the information-theoretic energy of the agent. By applying the Principle of Least Action to a Lagrangian density of Expected Free Energy ($\langle F \rangle$), we prove that:

1. The Information Energy Density (T_{00}) corresponds to the total predictive volatility, encompassing both current surprisal and future uncertainty.
2. The Information Momentum (T_{0i}) represents the directional flux of inference, quantifying how the agent updates its internal model to resolve ambiguity.
3. The Interaction Stress (T_{ij}) accounts for the structural resistance to state transitions, providing a formal basis for behavioral inertia and market frictions.

By coupling this predictive source tensor to the Ricci curvature ($R_{\mu\nu}$), we provide a unified field equation that describes how the "mass" of uncertainty—decomposed into Ambiguity and Relative Risk—curves the "space" of behavior. In this framework, the agent's actions are not simply responses to stimuli but are geodesic trajectories on a manifold whose curvature is perpetually updated by the stress-energy of perceived and expected information. This paper details the mathematical proof of this coupling and explores its implications for the study of complex adaptive systems, offering a structural solution to the exploration-exploitation trade-off through the lens of Riemannian geometry.

2 The Expected Free Energy Action Functional

To define the dynamics of an interaction manifold $\mathcal{M}$, we consider an agent seeking to minimize its future uncertainty. We define the Expected Free Energy ($\langle F \rangle$) as the path-integral of surprisal over predicted states. Following the expansion $\langle F \rangle = \sum_i p_i \mathcal{F}_i$, we express the information-theoretic Lagrangian density $\mathcal{L}_{\text{inf}}$ as:

$$\mathcal{L}_{\text{inf}}(\langle F \rangle, \nabla \langle F \rangle) = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle - V(\langle F \rangle).$$

Following Friston's principle of active inference, the expected free energy for a given policy or path can be decomposed into two fundamental information-theoretic quantities: Ambiguity and Relative Risk.

$$\begin{aligned} \langle F \rangle &= \underbrace{E_{q(s,o)}[\ln q(s)]}_{\text{Information}} - \underbrace{\ln p(s,o)}_{\text{Entropy}} \\ \langle F \rangle &\approx \underbrace{H(p(o|s))}_{\text{Ambiguity}(\mathcal{A})} + \underbrace{D_{KL}[q(s)||p(s)]}_{\text{Relative Risk}(\mathcal{R})} \end{aligned}$$

$$\langle F \rangle \approx \mathcal{A} + \mathcal{R}$$

Here, $\mathcal{A}$ denotes Ambiguity (entropy of the likelihood) and $\mathcal{R}$ denotes Relative Risk (Kullback-Leibler divergence from preferred states).

$V(\langle F \rangle)$ represents the potential landscape of the agent's long-term objectives.

The total action S of the system is the sum of the geometric (Einstein-Hilbert) action and the information action:

$$S = \int_{\mathcal{M}} \left[\frac{1}{2\kappa} (R - 2\Lambda) + \mathcal{L}_{\text{inf}} \right] \sqrt{-g} d^3x$$

3 Proof of the Interaction Source Tensor $T_{\mu\nu}$

Applying the principle of least action by varying the functional with respect to the inverse metric $g^{\mu\nu}$:

$$\frac{\delta S}{\delta g^{\mu\nu}} = 0 \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

The derivation of $T_{\mu\nu}$ from the EFE Lagrangian yields:

$$T_{\mu\nu} = \nabla_{\mu} \langle F \rangle \nabla_{\nu} \langle F \rangle - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \nabla_{\alpha} \langle F \rangle \nabla_{\beta} \langle F \rangle + V(\langle F \rangle) \right)$$

By substituting $\langle F \rangle \approx \mathcal{A} + \mathcal{R}$, we obtain the full expanded form:

$$T_{\mu\nu} = \nabla_{\mu} (\mathcal{A} + \mathcal{R}) \nabla_{\nu} (\mathcal{A} + \mathcal{R}) - g_{\mu\nu} \left(\frac{1}{2} |\nabla(\mathcal{A} + \mathcal{R})|^2 + V(\mathcal{A} + \mathcal{R}) \right)$$

4 Comprehensive Analysis of $T_{\mu\nu}$ Components

The tensor $T_{\mu\nu}$ represents the "Information Stress-Energy" that curves the interaction manifold. Its components provide a structural decomposition of behavioral drivers:

4.1 Information Energy Density (T_{00})

$$T_{00} = \frac{1}{2} \left[(\dot{\mathcal{A}} + \dot{\mathcal{R}})^2 + |\nabla(\mathcal{A} + \mathcal{R})|^2 \right] + V(\langle F \rangle)$$

T_{00} quantifies the Predictive Volatility of the system. It acts as the "effective mass" of the interaction. Regions of high ambiguity (high $\mathcal{A}$) or significant goal-deviation (high $\mathcal{R}$) increase the local curvature, effectively creating a "gravity of uncertainty" that pulls the agent's trajectory toward more stable or rewarding states.

4.2 Information Momentum and Learning Flux (T_{0i}, T_{i0})

$$T_{0i} = (\dot{\mathcal{A}} + \dot{\mathcal{R}})\nabla_i(\mathcal{A} + \mathcal{R})$$

These off-diagonal components represent the Information Flow Rate. They describe how the agent updates its internal model in response to shifting risks and ambiguities. In socioeconomic contexts, this corresponds to the "momentum" of a decision—how fast an agent moves to resolve uncertainty in the face of perceived risk.

4.3 Interaction Stress and Anisotropic Pressure (T_{ij})

$$T_{ij} = \nabla_i(\mathcal{A} + \mathcal{R})\nabla_j(\mathcal{A} + \mathcal{R}) - g_{ij}\mathcal{L}_{\text{inf}}$$

This represents the Structural Resistance within the interaction space. High values of T_{ij} indicate "Information Pressure," where the manifold resists certain state transitions due to the combined presence of high ambiguity and high risk. This provides a formal explanation for phenomena like "market friction" or "behavioral inertia," where agents remain trapped in suboptimal states due to the extreme geometric stress of moving through high-uncertainty regions.

5 Analysis of the Interaction Field Equations

The complete field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ reveals a self-correcting dynamical system:

1. Geometric Response to Agency: The Ricci tensor $R_{\mu\nu}$ represents the realized behavioral patterns. The equation states that behavioral curvature is directly proportional to the stress of expected surprisal. If the agent's expectations (EFE) are high, the "behavioral space" becomes highly curved, narrowing the set of probable actions.
2. The Role of Λ (Behavioral Attractor): The cosmological constant Λ acts as the Intrinsic Drive toward NESS (Nonequilibrium Steady-State). It ensures that even in the absence of external stimuli, the system maintains a baseline "homeostatic" curvature.
3. Bayesian Feedback Loop: Because the metric $g_{\mu\nu}$ is state-dependent, a change in behavior (curvature) updates the agent's position in the state-space, which in turn alters the EFE ($\langle F \rangle$) and the source tensor $T_{\mu\nu}$. This creates a closed-loop system where perception and action are coupled through the medium of Riemannian geometry.

6 Predictive Interaction Field Equation

From a conservative (Einstein's form) field equation in a condition of interaction source tensor,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Expand an interaction source tensor $T_{\mu\nu}$, by a principle of EFE Lagrangian,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(\nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle - g_{\mu\nu} \left(\frac{1}{2}g^{\alpha\beta} \nabla_\alpha \langle F \rangle \nabla_\beta \langle F \rangle + V(\langle F \rangle) \right) \right)$$

Arrange by using a group of $g_{\mu\nu}$, so that,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} + \kappa g_{\mu\nu} \left(\frac{1}{2}g^{\alpha\beta} \nabla_\alpha \langle F \rangle \nabla_\beta \langle F \rangle + V(\langle F \rangle) \right) = \kappa (\nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle)$$

$$R_{\mu\nu} - \left(\frac{1}{2}R - \Lambda - \kappa \frac{1}{2}g^{\alpha\beta} \nabla_\alpha \langle F \rangle \nabla_\beta \langle F \rangle + V(\langle F \rangle) \right) g_{\mu\nu} = \kappa (\nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle)$$

Let $\Phi = \frac{1}{2}R - \Lambda - \kappa \frac{1}{2}g^{\alpha\beta} \nabla_\alpha \langle F \rangle \nabla_\beta \langle F \rangle + V(\langle F \rangle)$, hence

$$R_{\mu\nu} - \Phi g_{\mu\nu} = \kappa \nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle$$

This equation is called a "Predictive Interaction Field Equation". Φ can be defined as a "Cumulative Structural Scalar Field", a scalar field of anisotropic scaling. It represents a structural background of a metric tensor $g_{\mu\nu}$ and controls it respectively. Furthermore, this equation describes a human behavior is controlled by a metric tensor, $g_{\mu\nu}$ (relies on a condition of $\nabla_{I_S}\gamma \neq 0$) directly. Responsively, a factor of $\kappa \nabla_\mu \langle F \rangle \nabla_\nu \langle F \rangle$ would rather supply a strength of $g_{\mu\nu}$ stiffness.

7 Conclusion

Through the derivation of the Interaction Source Tensor from the Expected Free Energy, we have established a rigorous field-theoretic basis for predictive agency. This framework demonstrates that the anomalies often cited in behavioral economics—such as risk-aversion and ambiguity-avoidance—are not irrational biases but are the necessary geometric consequences of minimizing information-theoretic stress. By treating $T_{\mu\nu}$ as a Bayesian Stress-Energy tensor, IFT provides a universal language for modeling any system that must infer its environment to survive. Future work will investigate the quantization of this interaction field to address stochasticity in micro-scale decision processes.